Romila Pradhan

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Prepared June 1, 2026

RESEARCH INTERESTS

The design of trustworthy data-driven decision-making systems, and the application of data management principles to high quality, trusted end decisions. Research techniques draw on data quality, data integration, data science pipelines, and machine learning fairness and robustness.

EDUCATION

2018 **Purdue University**, West Lafayette, IN
Ph.D., Computer Science
Advisor: Sunil Prabhakar
Dissertation: *Guided Data Fusion*

2008 **Indian Institute of Technology**, Kharagpur, India
M.S., Mathematics and Computing
Advisor: M. P. Biswal
Thesis: *Multiple Choice Linear Programming – The Problem and the Solutions*

2008 **Indian Institute of Technology**, Kharagpur, India
B.S., Mathematics and Computing
Advisor: G. P. Rajasekhar
Dissertation: *Branch-Checking Algorithm for All-Pairs Shortest Paths*

PROFESSIONAL EXPERIENCE

2021–current **Purdue University**, West Lafayette, IN
Assistant Professor – School of Applied and Creative Computing
Affiliate Faculty Member – Center for Education and Research in Information Assurance and Security (CERIAS)
Core Faculty Member – Applied AI Research Center

2020–2021 **University of California San Diego**, La Jolla, CA
Postdoctoral Researcher – Halıcıoğlu Data Science Institute

2018-2020 **Purdue University**, West Lafayette, IN
Visiting Assistant Professor – Department of Computer Science

2011–2018 **Purdue University**, West Lafayette, IN
Ph.D. Student – Department of Computer Science

2008–2010 **Deutsche Bank Group**, Mumbai, India
Senior Analyst – Interest Rates Group

AWARDS

2026 EDBT Conference Best Reviewers Award
Outstanding Faculty in Discovery, School of Applied and Creative Computing
Exceptional Early Career Teaching Award, School of Applied and Creative Computing

2023 NSF CAREER Award

2022 Google Research Scholar Award

GRANTS

2026 NSF REU: CAREER: *Data Preparation for Trusted and Fair Data Science*
PI: Romila Pradhan

2023 NSF CAREER: *Data Preparation for Trusted and Fair Data Science*
PI: Romila Pradhan

Center for Advancing Safety of Machine Intelligence (CASMI): *Diagnosing, Understanding, and Fixing Data Biases for Trusted Data Science*
PI: Romila Pradhan

2022 Google Research Scholar Award
PI: Romila Pradhan

PUBLICATIONS

Authorship Conventions

Conferences are the primary publication venue in databases and data management. Senior authors are typically listed last. Advisees are listed in **bold**. Romila Pradhan is underlined.

In Progress or In Review

2026 [1] **A. Singh**, R. Pradhan. *Scalable Source Selection for Machine Learning Tasks.* Under revision.

[2] **J. Hasan**, **J. Wang**, R. Pradhan. *Explanations and Recommendations for Machine Learning Pipelines under Data Drift.* In Progress.

[3] **J. Wang**, L. Lin, R. Pradhan. *Source-Free Tabular Feature Unlearning via Selective Knowledge Distillation and Conditional Tabular Refinement.* In Review.

Conference Proceedings

2026 [4] **J. Hasan**, S. Jiang, **T. Singh**, S. Galhotra, R. Pradhan, D. Srivastava. *PipeLens: Identifying Interventions for Resolving Malfunctioning Data Science Pipelines.* VLDB 2026.

[5] **A. Singh**, R. Pradhan. *Scalable Goal-Oriented Source Selection.* QDB Workshop @VLDB 2026.

[6] **J. Hasan**, R. Pradhan. *Selective Data Expansion for Model Performance.* EDBT 2026.

[7] **J. Hasan**, **J. Wang**, R. Pradhan. *SENTINEL: Evaluating Pipeline Robustness to Distributional Shifts.* ICDE 2026. (Demonstration).

[8] **A. Singh**, R. Pradhan. *SPLICE: An Efficient Framework for Selecting Sources for Machine Learning Tasks.* ICDE 2024. (Lightning talk).

[9] **J. Hasan**, **J. Wang**, R. Pradhan. *Evaluating Pipeline Robustness Under Distributional Shifts.* ICDE 2024. (Lightning talk).

2025 [10] **T. Surve**, R. Pradhan. *Explaining Fairness Violations using Machine Unlearning.* EDBT 2025.

[11] **S. Thandri**, R. Pradhan. *Label Flipping for Group Fairness.* QDB Workshop @VLDB 2025.

[12] **J. Hasan**, R. Pradhan. *Explanations for Machine Learning Pipelines under Data Drift.* HILDA Workshop @SIGMOD 2025.

2024 [13] **Ekta**, R. Pradhan. *Valuation-based Data Acquisition for Machine Learning Fairness.* QDB Workshop @VLDB 2024.

2022 [14] R. Pradhan, T. Li. *Human-in-the-loop Bias Mitigation in Data Science.* HCAI Workshop @ NeurIPS 2022.

[15] R. Pradhan, J. Zhu, B. Glavic, B. Salimi. *Interpretable Data-based Explanations for Fairness Debugging.* SIGMOD 2022.

[16] J. Zhu, R. Pradhan, B. Glavic, B. Salimi. *Generating Interpretable Data-Based Explanations for Fairness Debugging using Gopher.* SIGMOD 2022 (Demonstration).

[17] R. Pradhan, A. Lahiri, S. Galhotra, B. Salimi. *Explainable AI: Foundations, Applications, Opportunities for Data Management Research.* SIGMOD 2022 (Tutorial).

[18] R. Pradhan, A. Lahiri, S. Galhotra, B. Salimi. *Explainable AI: Foundations, Applications, Opportunities for Data Management Research.* ICDE 2022 (Tutorial).

2021 [19] S. Galhotra*, R. Pradhan*, B. Salimi. *Explaining Black-Box Algorithms Using Probabilistic Contrastive Counterfactuals.* SIGMOD 2021. (*=Equal contribution)

[20] S. Galhotra, R. Pradhan, B. Salimi. *Feature Attribution and Recourse via Probabalistic Contrastive Counterafctuals.* Algorithmic Recourse Workshop @ICML 2021.

[21] P. Y. Wang, S. Galhotra, R. Pradhan, B. Salimi. *Demonstration of Generating Explanations for Black-Box Algorithms Using Lewis.* VLDB 2021 (Demonstration).

2019 [22] R. Pradhan, S. Prabhakar. *AuthIntegrate: Toward Combating False Data on the Internet.* Truth Discovery and Fact Checking Workshop @SIGKDD 2019.

2018 [23] R. Pradhan, S.Bykau, S. Prabhakar. *A Framework to Integrate User Feedback for Rapid Conflict Resolution.* ICDE 2018 (Demonstration).

[24] R. Pradhan, W. G. Aref, S. Prabhakar. *Leveraging Data Relationships to Resolve Conflicts from Disparate Data Sources.* DEXA 2018.

2017 [25] R. Pradhan, S. Bykau, S. Prabhakar. *Staging User Feedback toward Rapid Conflict Resolution in Data Fusion.* SIGMOD 2017.

TEACHING EXPERIENCE

Course Development

2022 Responsible Data Management, Purdue CNIT graduate course

2019 Large-scale Data Analytics, Purdue CS undergraduate course

Teaching

2026 Spring Instructor, *Enterprise Data Management* (CNIT 39200, 81 students)

2025 Fall Instructor, *Responsible Data Science* (CNIT 57100, 19 students)

	Instructor, *Database Administration* (CNIT 48700, 7 students)
2025 Spring	Instructor, *Enterprise Data Management* (CNIT 39200, 81 students)
2024 Fall	Instructor, *Responsible Data Science* (CNIT 57100, 7 students)
	Instructor, *Database Administration* (CNIT 48700, 14 students)
2024 Spring	Instructor, *Enterprise Data Management* (CNIT 39200, 61 students)
	Instructor, *Responsible Data Management* (CNIT 58100-RDM, 13 students)
2023 Fall	Instructor, *Responsible Data Science* (CNIT 58100-RDM, 19 students)
	Instructor, *Database Administration* (CNIT 48700, 7 students)
2023 Spring	Instructor, *Enterprise Data Management* (CNIT 39200, 56 students)
	Instructor, *Responsible Data Management* (CNIT 58100-RDM, 12 students)
2022 Fall	Instructor, *Responsible Data Science* (CNIT 58100-RDM, 5 students)
	Instructor, *Database Administration* (CNIT 48700, 5 students)
2022 Spring	Instructor, *Enterprise Data Management* (CNIT 39200, 61 students)
2022 Fall	Instructor, *Database Administration* (CNIT 48700, 6 students)

STUDENTS

Graduate Ph.D.s

Ikechukwu Andrew Ben Obi, (co-chaired with B. C. Min). Expected 2026
Neurosymbolic Robot Learning for Safe Language-Controlled Robot Systems

Current Ph.D.s

Jahid Hasan. 2023–current
Intelligent Data-Centric Systems for Fair, Robust, and Adaptive Data Science Pipelines

Ambarish Singh. 2023–current
Efficient Source Selection for Machine Learning

Jingya Wang. 2025–current, *Andrews Fellowship*
Toward Practical Machine Unlearning Techniques

M.S. Advising

Omkar Pote. 2024–current
Optimizing Machine Learning Pipelines Under Data Drift

Ananya Uppal. 2024–2026
Role of Data Valuation Techniques in Detecting and Repairing Data Quality Issues in Machine Learning Pipelines

Harshita Rathee. 2024–2026
The Role of Data Selection on the Fairness of Large Language Models

Kevin Chittilapilly. 2023–2024
Estimating Fairness in Machine Learning Systems through Data Characteristics

Tejendra Singh. 2023–2024
Leveraging Data Characteristics to Optimize Machine Learning Pipelines

Shashank Thandri. 2023–2024
Improving Fairness in Machine Learning Systems through Label Flipping

Ekta 2022–2024.
Using Data Acquisition to Improve Machine Learning Model Fairness

Sathvika Kotha. 2022–2024
Studying the Impact of Data Quality and Preprocessing on ML Model Fairness

Tanmay Surve. 2022–2023
Data-Based Explanations of Random Forests using Machine Unlearning

Ph.D. Committees

Weizhang Wang. 2025–current
Reinforcement Learning from Human Feedback and Large Language Model Logical Reasoning

Srujana Kaddevarmuth. 2024–current
Comprehensive Ethical Frameworks for Developing and Scaling AI Systems

James Prater. Defended 2025
Effects of Automation Bias on Human-AI Teaming in High-stress Environments

Tung Vu. Defended 2023
A Statistical Approach to Detecting Generative Adversarial Network Overfitting

M.S. Committees

Dhairya Dhuvad. 2025–current
Fairness of Locally Differentially Private Models

Meghana Gorripati. 2025-2026
Explainable AI for Enhancing Cloud Autoscaling

Rohit Bankar. 2025
Resource Optimization for Cloud-native ML Models

Athreyan Mohana. 2024-2025
Ranked Cluster Influence: A Defense against Adversarial Labeling Attacks on Fairness

Shridhar Ayush. 2023-2024
Intelligent Auto-scaling of Cloud-native Machine Learning Workloads

Anuj Dubey. 2023-2024
Capturing Style through Large Language Models: An Authorship Perspective

Yiqun Zhang. 2023
Predicting Answer Acceptance on Stack Overflow Using Asker's Participation in Answer Comments

Kanwardeep Singh Walia. 2021-2022
Using Temporal Networks to Find the Influencing Buggy Site in the Code Communities

External Committees

Haakon Robinson, Ph.D., Norwegian University of Science and Technology. Defended 2023
Trustworthy Machine Learning for Controlled Dynamical Systems

Aksel Vaaler, M.S., Norwegian University of Science and Technology. Defended 2023
Design and Simulation of Predictive Safety Filter for Autonomous Ship

Undergrad Research Advising

Bo-Cheng Lu. 2026-current
Ensuring Fairness in Machine Learning Pipelines

Ritwik Jayaraman. 2025-current
Zero-shot Machine Unlearning for Tabular Data

Animesh Joshi. 2024-2025
Debugging Machine Learning Pipelines

Ahana Bhattacharyya (University of Buffalo). 2023-2024, Ph.D. at Brown University
Data Validity for Machine Learning Model Fairness

Liming Liu. 2023

Anusha Sarraf. 2023
Investigating Explainability of Machine Learning Model Decisions for Injury Surveillance

Manas Paranjape. 2022
Investigating Explainability of Machine Learning Model Decisions for Injury Surveillance

Andrea Sanchez. 2022
Detecting Bias in ML-based Credit Risk Algorithms

Jiawei Chang. 2022
Detecting Bias in ML-based Income Prediction

Jiongli Zhu. 2022
Fairness Debugging via Interpretable Data-based Explanations

Rohith Ravindranath. 2019
Identifying Direct and Indirect Causal Effects on Outcomes in Healthcare

INVITED TALKS

2025 Explainability, Robustness, and Fairness of Data Science Applications
School of Applied and Creative Computing Research Seminar Series, Purdue University

Debugging, Explaining, and Mitigating Unfairness in Machine Learning
Cornell University

2024 Explaining Black-box Probabilistic Attacker-Defender Models
Sandia National Laboratories

Debugging and Explaining Unfairness in Machine Learning Models
Applied Artificial Intelligence Research Center, Purdue University

Debugging and Explaining Unfairness in Machine Learning Models
CERIAS Symposium, Purdue University

Diagnosing, Understanding, and Fixing Data Biases for Trusted Data Science
Center for Advancing Safety in Machine Intelligence

2023 Explainability and Fairness in Algorithmic Systems
Honors College, Purdue University

Toward Trustworthy and Responsible Data Science
Brandeis University

Interpretable Data-based Explanations for Fairness Debugging
Industrial Engineering, Purdue University

Causality-based Post hoc Explanations for Fairness Debugging
Computer Science & Artificial Intelligence Laboratory, MIT

2022 Data Preparation for Fair and Trusted Machine Learning
Cisco Research

2021 Explaining Black-box Algorithms Using Probabilistic Contrastive Counterfactuals
Computer Science & Artificial Intelligence Laboratory, MIT

2021 Explaining Black-box Algorithms Using Probabilistic Contrastive Counterfactuals
Department of Computer Science, University of California San Diego

UNIVERSITY SERVICE

2021-current Data and Software Development Curriculum Sub-Committee
School of Applied and Creative Computing

2022-2026 Department Faculty Search Committee
School of Applied and Creative Computing

2025-2026 Co-chair, Research Synergies Working Group
School of Applied and Creative Computing

2025 Organizer, Applied AI Day
Purdue Polytechnic Institute

2022-current Library Committee, Purdue University

COMMUNITY SERVICE

2026 SIGMOD PC
VLDB PC
EDBT PC
ICDE PC
TKDE Review Board
TWeb Review Board
TDSC Review Board
TII Review Board
HILDA PC and Mentor
QDB PC

2025 SIGMOD PC
VLDB PC
EDBT PC
ICDE PC
TaDA PC
TKDE Review Board
TWeb Review Board
TDSC Review Board
NASEM Rapid Expert Consultation Reviewer

2024 EDBT PC
SDM PC
CIKM PC
TKDE Review Board
TWeb Review Board
TDSC Review Board
NSF IIS Panelist

2023 EDBT PC
SIGKDD PC
FAccT PC
CIKM PC
TKDE Review Board
TWeb Review Board
TDSC Review Board
ICDE PhD Symposium PC
NSF IIS Panelist
US-Israel Binational Science Foundation Panelist

2022	SIGMOD PC EDBT PC CIKM PC SIGKDD PC TKDE Review Board TDSC Review Board
2021	TKDE Review Board TDSC Review Board
2020	TKDE Review Board TDSC Review Board
2019	TDSC Review Board

OUTREACH

2024	VISION Summer camp course on Bias in Machine Learning *Purdue Polytechnic Institute*
	Discovering Opportunities in Technology (DOiT) camp course on Bias in Machine Learning *Purdue University*
2022	Hands-on activities on Machine Learning and Bias to students from Purdue Polytechnic High School *Purdue University*
	Summer camp data science module on Machine Learning and Bias *Purdue University*
2019	Summer camp data science module on exploratory data analysis and classification to Global Engineering Program participants from Taiwan *Purdue University*

Glossary

SIGMOD	ACM SIGMOD/PODS International Conference on Management of Data
VLDB	International Conference on Very Large Data Bases
ICDE	IEEE International Conference on Data Engineering
EDBT	International Conference on Extending Database Technology
HILDA	Human-in-the-Loop Workshop at SIGMOD
QDB	Quality in Databases Workshop at VLDB
TaDA	Tabular Data Analysis Workshop at VLDB
TKDE	IEEE Transactions on Knowledge and Data Engineering
TWeb	IEEE Transactions on Web
TDSC	IEEE Transactions on Dependable and Secure Computing
TII	IEEE Transactions on Industrial Informatics
NSF	National Science Foundation
III	Division of Information and Intelligent Systems at NSF
REU	Research Experience for Undergraduates
NASEM	National Academies of Sciences, Engineering, and Medicine